

RIDEOPS

Driver Retention & Performance Intelligence

Business Intelligence Portfolio Case Study

1. Executive Summary

RideOps is an end-to-end Business Intelligence solution designed to analyze driver retention, performance deterioration, and operational risk in a ride-hailing environment. The project transforms longitudinal driver data into an analytical model, performs business analysis using SQL and Python, and presents the results through four interactive Tableau dashboards.

The solution is structured around four operational questions: where is attrition occurring, when does retention deteriorate in the driver lifecycle, which drivers are showing signs of deterioration, and which drivers should an operations team prioritize for investigation.

The four-dashboard workflow is: Fleet Overview → Cohort & Lifecycle Survival → Driver Health → Retention Action Center.

Project at a Glance

Unique drivers	2,381
Monthly driver observations	19,104
Monthly churn in selected operational view	9.52%
Total GMV in selected operational view	\$571.9M
Month 12 survival	38.1%
Current At-Risk drivers	190
At-Risk business value	\$10.74M
High-value At-Risk drivers	80

2. Business Problem

Driver attrition can create operational instability and increase the need for driver acquisition. A simple churn report can show how many drivers left, but it does not explain when attrition occurs, which cohorts are weaker, which cities have greater exposure, or which active drivers are showing signs of deterioration.

RideOps therefore focuses on:

- Monitoring driver attrition over time.
- Analyzing driver lifecycle and cohort retention.
- Comparing retention across cities and joining designations.
- Tracking income, business value, rating, and grade movement.
- Creating an interpretable driver-health framework.
- Identifying active drivers requiring operational attention.
- Quantifying business value associated with at-risk drivers.
- Providing decision-ready dashboards for operations stakeholders.

3. Dataset

The project uses a publicly available OLA Driver Performance & Attrition dataset as a proxy for a ride-hailing operations environment. The dataset is longitudinal: the same driver can appear across multiple reporting months.

- 2,381 unique drivers.
- 19,104 monthly driver observations.
- Driver demographics and joining information.
- City and joining designation.
- Monthly income and total business value.
- Driver grade and quarterly rating.
- Last working date.

Scope note: The dataset is not Pathao data. Pathao is used as the target business context for the portfolio project; the analysis demonstrates the BI workflow rather than making claims about Pathao's internal operations.

4. Data & Analytics Architecture

The workflow can be summarized as:

Raw Data	Python / Pandas	Analytical Model	PostgreSQL / SQL	Tableau	Insights	Operational Action
----------	-----------------	------------------	------------------	---------	----------	--------------------

Analytical Tables

- `dim_driver` — relatively static driver attributes such as driver ID, city, education, joining date, and joining designation.
- `fact_driver_monthly` — time-varying attributes such as reporting date, income, business value, grade, rating, last working date, churn status, tenure, and derived performance changes.

5. Data Preparation & Feature Engineering

- Load and standardize the raw dataset.
- Parse and validate date fields.
- Create churn indicators.
- Calculate driver tenure.
- Generate previous-period metrics.
- Calculate month-over-month income, business-value, and rating changes.
- Construct the analytical driver and monthly fact tables.
- Create the operational Driver Health Score.

6. Driver Health Framework

RideOps uses an interpretable 0–100 Driver Health Score to prioritize operational investigation. The score combines several performance signals rather than relying on a single metric.

Component	Weight
Income trend	25%

Business-value trend	25%
Current rating	20%
Rating movement	15%
Tenure	15%

Score	Category
70–100	Healthy
50–69.99	Stable
35–49.99	Watch List
<35	At Risk

The score is a heuristic operational-prioritization framework, not a calibrated probability of churn.

7. Dashboard 1 – Executive Overview

The Executive Overview provides the fleet-level operational picture and connects churn with city-level business exposure.

- 819 drivers in the selected operational view.
- 9.52% monthly churn.
- \$571.9M total GMV.
- 4 cities above the dashboard's >75% churn threshold.
- Monthly attrition trend.
- City churn versus revenue exposure.
- City-level leaderboard.

Primary business question: Where is the retention problem occurring, and where does it carry the greatest operational exposure?

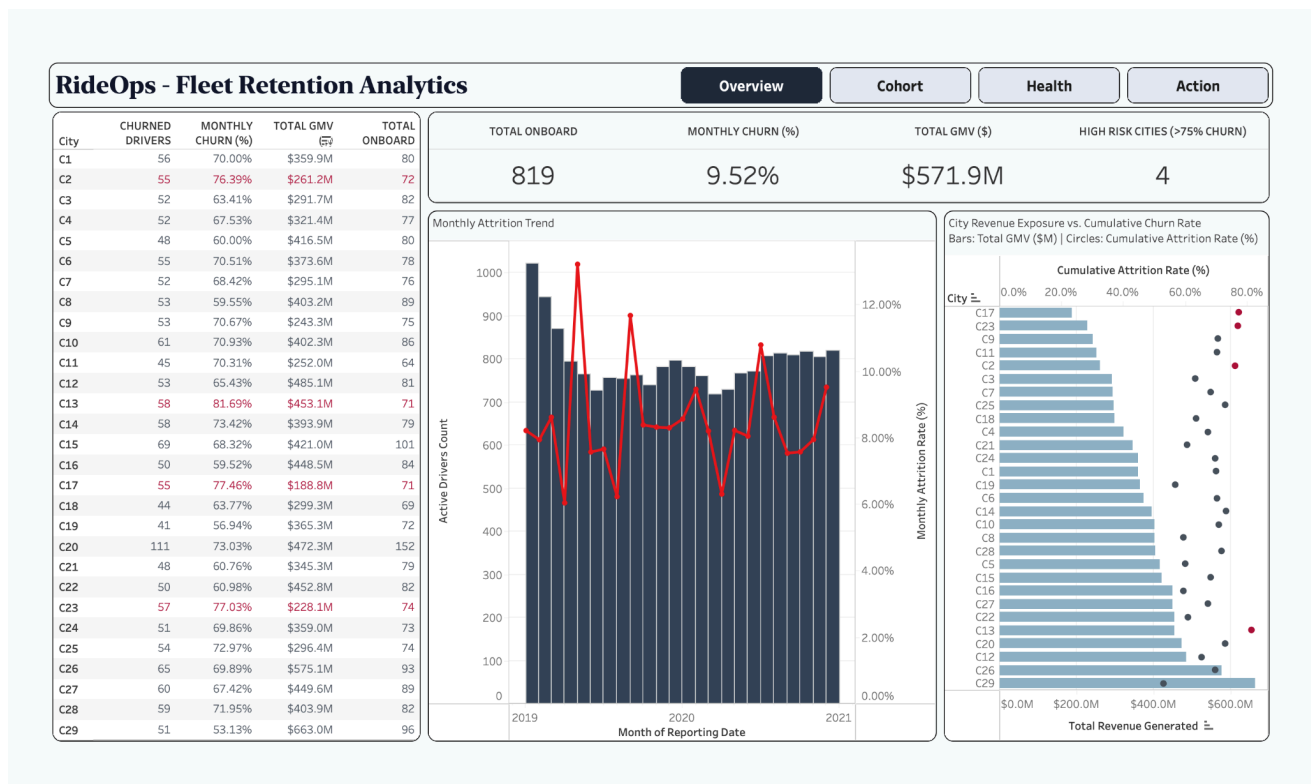


Figure 1. RideOps Executive Overview — fleet retention, churn, and city revenue exposure.

8. Dashboard 2 — Driver Cohort & Lifecycle Survival

The Cohort dashboard focuses on when drivers leave and how retention changes throughout the driver lifecycle.

Month 1 survival	96.6%
Month 3 survival	88.8%
Month 6 survival	68.0%
Month 12 survival	38.1%

The dashboard contains a cohort retention heatmap, survival curves by joining designation, and attrition comparisons by education level and grade.

Key finding: survival declines from 96.6% at Month 1 to 38.1% at Month 12, indicating that retention should be analyzed as a lifecycle problem rather than only as a month-to-month churn metric.

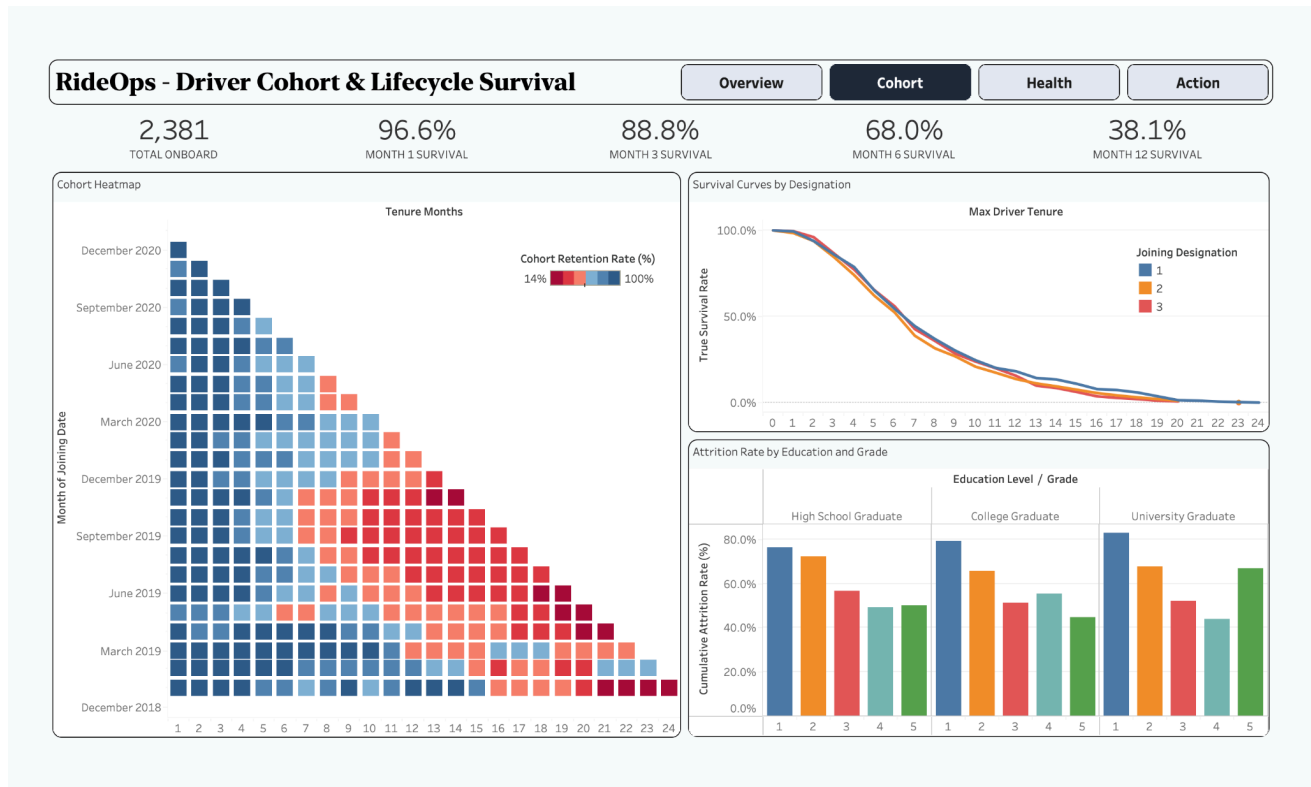


Figure 2. RideOps Cohort & Lifecycle Survival — cohort retention, survival curves, and attrition segmentation.

9. Dashboard 3 — Driver Health

The Health dashboard moves from fleet-level reporting to driver-level risk monitoring.

- Fleet Health Breakdown — current operational distribution across Healthy, Stable, Watch List, and At Risk.
- Decay Quadrant — Rating Shift versus Income Change (%).
- Health Migration Trend — health-category distribution across reporting months.

In the 819-driver operational view, the Fleet Health Breakdown shows: Watch List 40.29% (330), Stable 25.76% (211), At Risk 23.20% (190), and Healthy 10.74% (88).

Important reporting note: the KPI cards on this dashboard use a different aggregation/scope from the donut chart. The KPI values should therefore not be presented as unique-driver counts alongside the donut values.

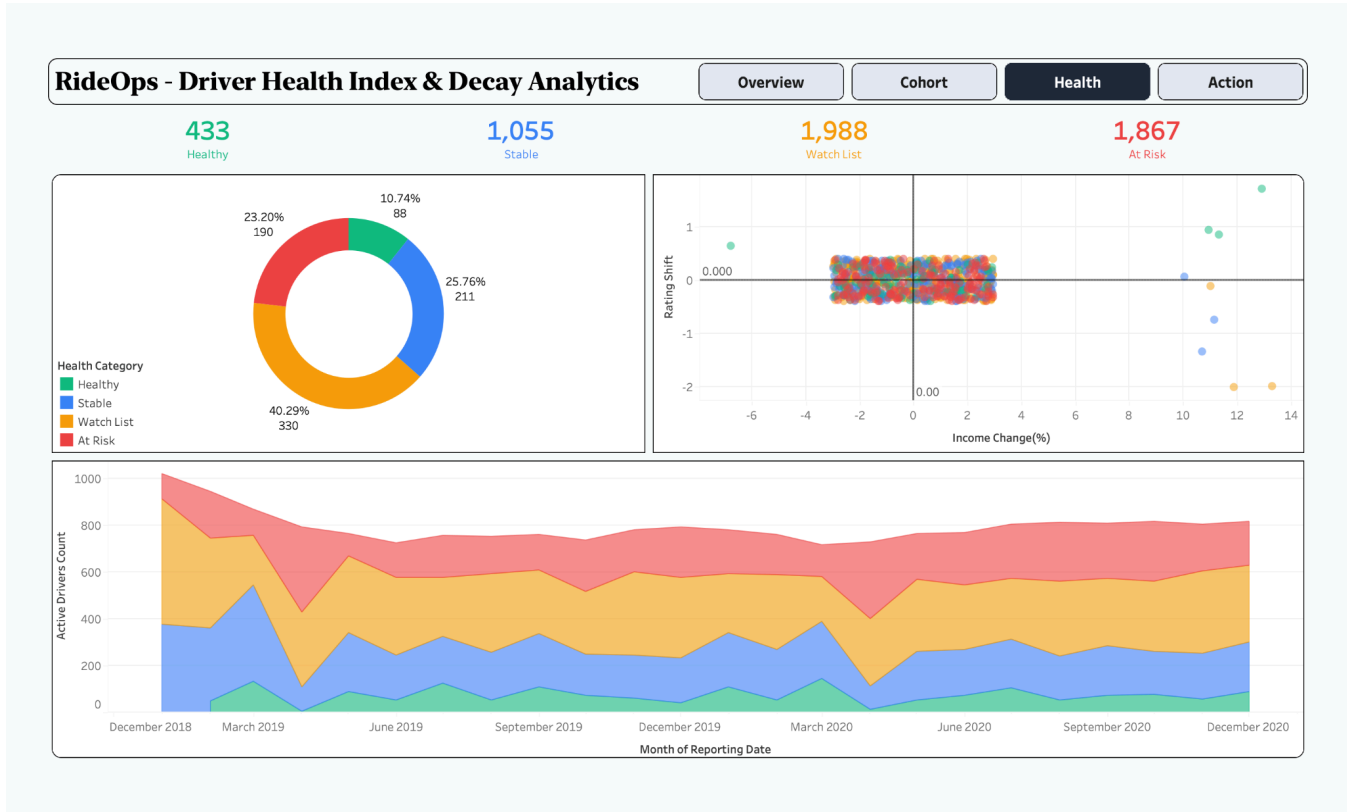


Figure 3. RideOps Driver Health — fleet health distribution, performance decay, and health migration

10. Dashboard 4 — Retention Action Center

The Action dashboard converts the analytical signals into an operational prioritization workflow.

At-Risk count	190
At-Risk business value	\$10.74M
Average income change	10.3%
High-value At-Risk	80

- Priority Intervention Roster — detailed profiles grouped by city with grade, education, health score, business value, income change, and average rating.
- Driver Profile Trend — individual health trajectories over time.
- Retention Sensitivity Curve — baseline versus simulated driver counts under a 10% incentive-uplift scenario.

The incentive analysis is exploratory scenario analysis and should not be interpreted as causal evidence that a 10% incentive will produce a specific retention improvement.

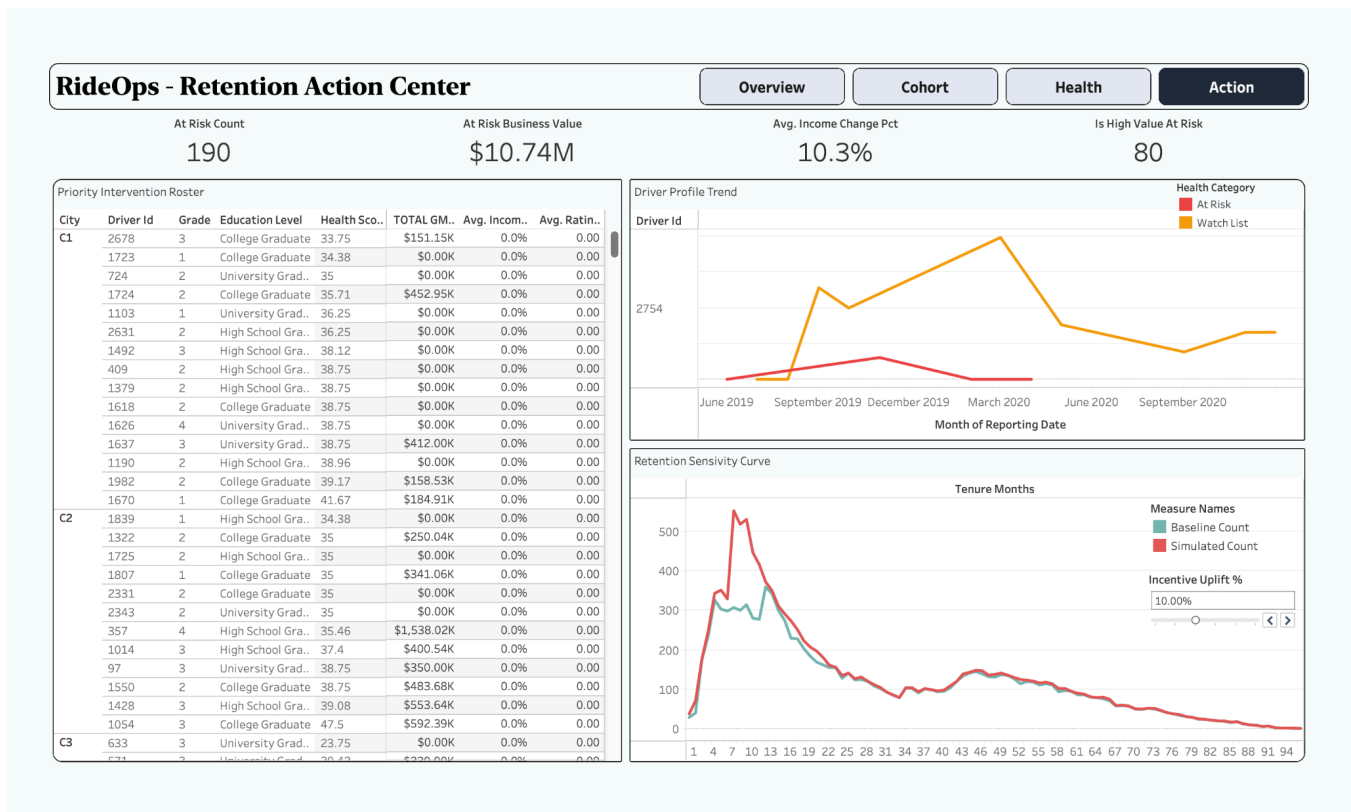


Figure 4. RideOps Retention Action Center — prioritized at-risk drivers and retention intervention analysis

11. Key Business Findings

Retention deteriorates substantially across the first year. Survival declines from 96.6% at Month 1 to 38.1% at Month 12, making driver lifecycle retention an important area for operational investigation.

Churn exposure varies by city. The Executive Overview shows meaningful differences in cumulative attrition and GMV across cities, so city-level retention should not be treated as uniform.

Churn should be considered alongside business exposure. The City Revenue Exposure view helps distinguish high churn from high churn combined with substantial GMV exposure.

A meaningful active population requires attention. The Action Center identifies 190 At-Risk drivers representing \$10.74M in business value, including 80 high-value At-Risk drivers.

Driver deterioration is multidimensional. Income and rating movement provide complementary signals, supporting a multi-factor health framework rather than a single-metric alert.

12. Business Recommendations

- Segment retention analysis by driver tenure and investigate the early-to-mid lifecycle decline.
- Prioritize operational review of At-Risk drivers with high business value.
- Monitor income, rating, and business-value deterioration together rather than using one metric in isolation.
- Investigate city-specific retention patterns before applying a uniform intervention across the network.

- Validate incentive or retention interventions through controlled experimentation before treating simulated uplift as an expected business outcome.

13. Limitations

- The project uses public OLA data rather than Pathao's internal data.
- The dataset does not contain individual ride-level transactions, online hours, passenger wait time, or ride-level cancellation behavior.
- The Driver Health Score is heuristic and has not been statistically calibrated as a churn probability.
- A missing LastWorkingDate does not prove permanent retention.
- The incentive-uplift analysis is hypothetical scenario analysis, not a causal estimate.

14. Future Improvements

- Integrate ride-level transactions, online hours, completed trips, and cancellation rates.
- Incorporate incentive history, acquisition channel, support interactions, and supply-demand metrics.
- Build a statistically calibrated churn model.
- Automate data refresh and Tableau publishing.
- Create automated alerts for newly emerging At-Risk drivers.
- Evaluate retention interventions using A/B testing or causal inference.

15. Conclusion

RideOps demonstrates an end-to-end BI workflow that moves from fleet-level monitoring to cohort analysis, driver-health monitoring, and operational action. The project connects raw longitudinal data to Python-based transformation, SQL analytics, Tableau visualization, and decision-oriented reporting.

Fleet → Cohort → Health → Action